Five Number of Summary:-

1. Minimum
2. First Quartile (25%) Q_1
3. Median (Q_2)
4. third Quartile (75%) Q_3
5. Maximum

form the given exp $D = [2, 4, 4, 4, 5, 5, 7, 9]$

minimum = 2

$Q_1 = 4$

$Q_2 = 4.5$

$Q_3 = 6$

maximum = 9

$\therefore IQR = Q_3 - Q_1 \Rightarrow 6-4$

$= 2$

Removing the outlier

[Lower fence ⟷ Higher fence]

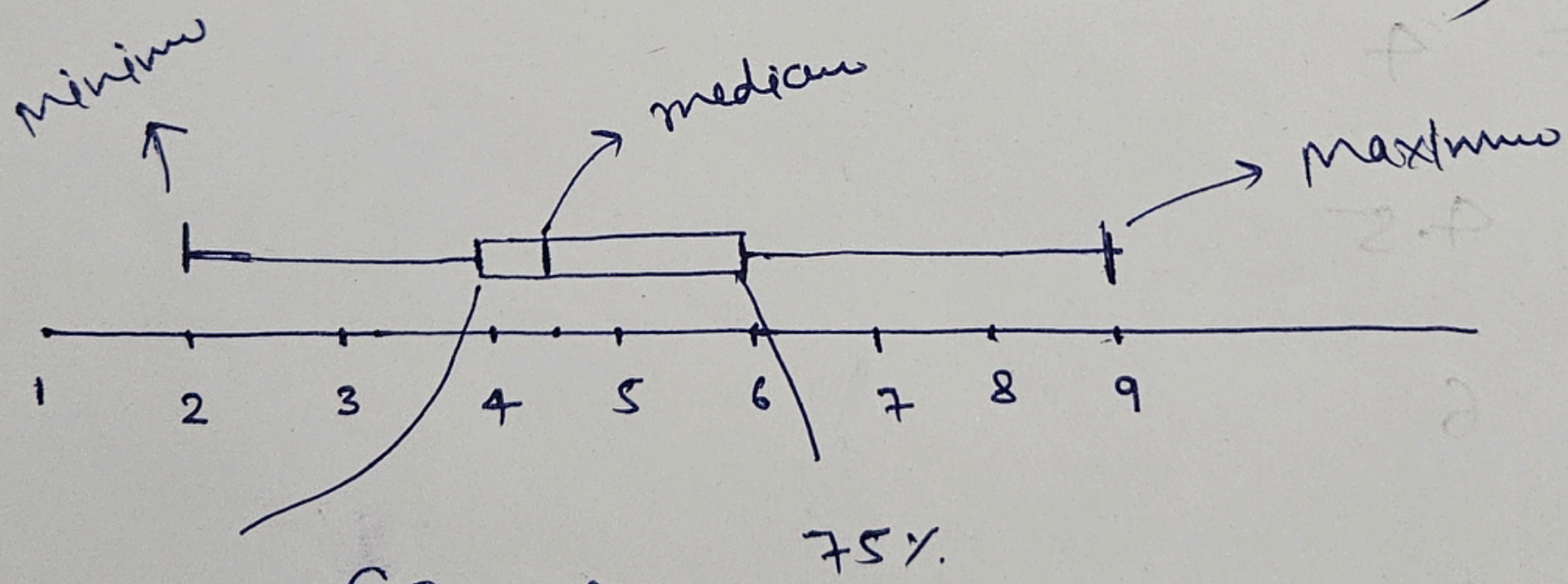
Lower fence = $Q1 - 1.5(IQR)$

Higher fence = $Q_3 + 1.5(IQR)$

Lower fence = $4 - 1.5 * 2 = 1$

upper fence = $6 + 1.5 * 2 = 9$

Data points outside (<1 and >9)



(Boxplot

IQR = Inter Quartile Range.

Q TO Calculate Five Number of Summary:-

Dataset = [7, 8, 5, 12, 14, 21, 13, 18, 20, 25]

Step(1) Arrangt in Ascending order

D = [5, 7, 8, 12, 13, 14, 18, 20, 21, 25]

Step(2)

$n = 10$

Find the five-Number Summary

minimum = 5

maximum = 25

median (Q_2): since 10 value (even)

median = average of 5th & 6th value

= (13+14)/2 = 13.5

Q1 (lower Quartile) → median of first half (5, 7, 8, 12, 13) = 8

Q_3 (upper Quartile) → median of second half

(14, 18, 20, 21, 25) = 20

Five Number Summary: -

Q1 = 5

maximum = 25

Q_1 = 8

Q_3 = 20

Q_2 = 13.5

(5, 8, 13.5, 20, 25)

Boxplot measurements :-

Interquartile range (IQR) = $Q_3 - Q_1$

= 20 - 8 = 12

lower fence = $Q_1 - 1.5 * IQR = 8 - 18 = -10$

Upper fence = $Q_3 + 1.5 * IQR = 20 + 18 = 38$

Any value below -10 or above 38 is an outlier
Here 25, in inside, but if dataset had 40, or 45 then we call

Calculate Five Number of summary using following dataset.

D = [3, 7, 8, 5, 12, 14, 21, 13, 18]

n = 9.

Step(1) Arrange in ascending order

D = [3, 5, 7, 8, 12, 13, 14, 18, 21]

Find the value

minimum = 3

maximum = 21

median (Q2) = middle value (5th value in 9 data points)

= 12

Q1 (Lower Quartile) = median of the lower half (3, 5, 7, 8)

= (5+7)/2 = 6

Q3 (Upper Quartile) = median of the upper half (13, 14, 18, 21)

= (14+18)/2 = 16

IQR = Q3 - Q1

= 16 - 6

= 10

Lower Fence = Q1 - 1.5 × IQR

= 6 - 1.5 × 10

= 6 - 15

= -9

Upper Fence = Q3 + 1.5 × IQR

= 16 + 1.5 × 10

= 16 + 15

= 31

Since all values lie between -9 and 31, there are no outliers.

Q_3 (upper Quartile)

= median of upper half (13, 14, 18, 21)

= (14+18)/2 = 16

Five Number of Summary

minimum = 3

Q_1 = 6

median (Q_2) = 12

Q_3 = 16

maximum = 21

outlier detection using Boxplot.

D = [7, 8, 5, 12, 14, 21, 13, 18, 20, 25, 45]

Step (1) : Arrange in Ascending order

D = [5, 7, 8, 12, 13, 14, 18, 20, 21, 25, 45]

n = 11

Step (2): Five Number Summary =

min : 5

Q_1 = 8 (median of lower half 5, 7, 8, 12, 13)

median (Q_2) = 14 (middle value)

Q_3 = (median of upper half : 14, 18, 20, 21, 25, 45 →)

(20+21)/2 = 20.5

max = 45

Boxplot calculations:-

$IQR = Q_3 - Q_1 = 21 - 8 = 13$

Lower fence $= Q_1 - 1.5 * IQR$

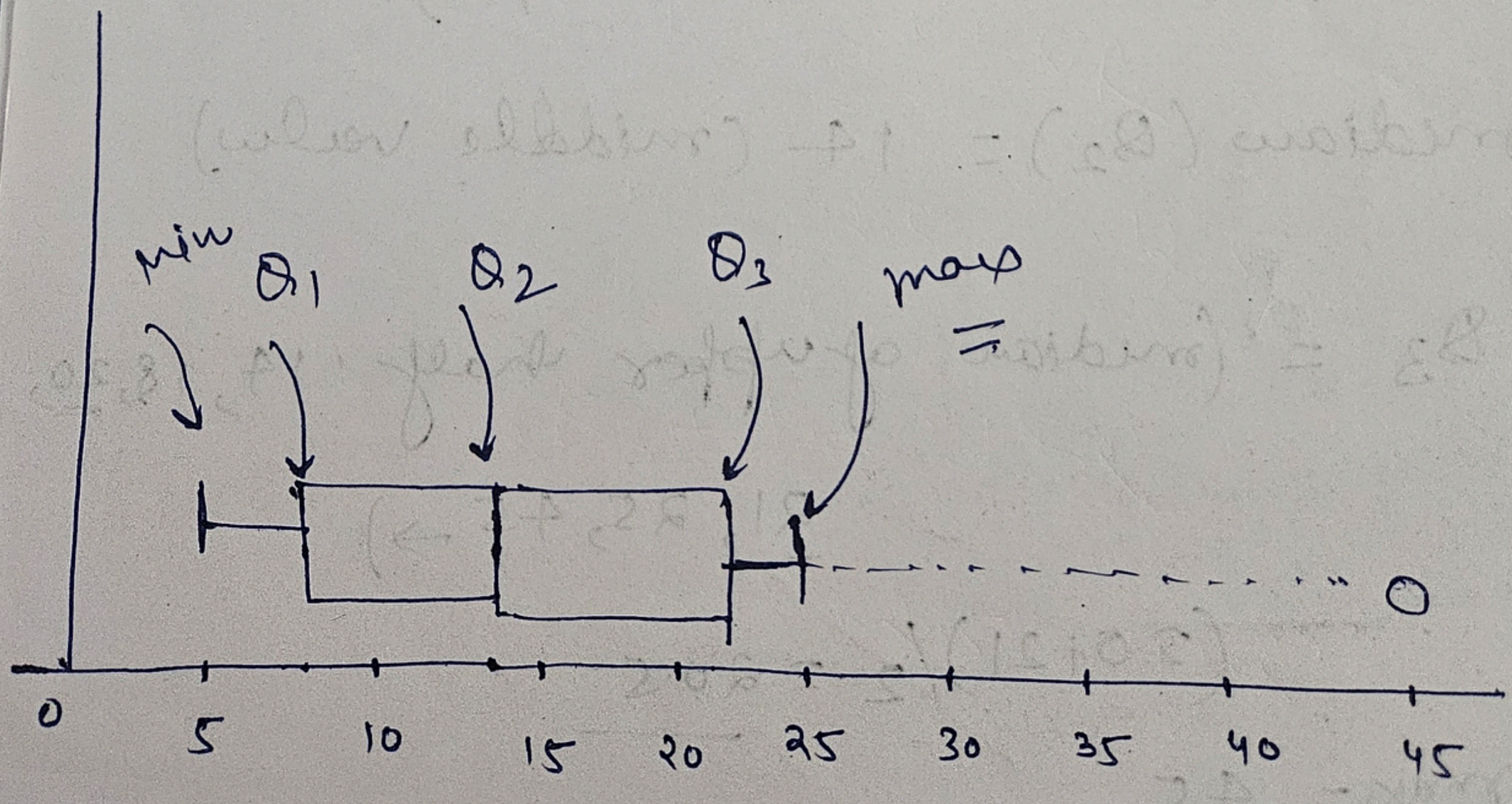
$= 8 - 19.5 = -11.5$

Upper fence $= Q_3 + 1.5 * IQR$

$= 20.5 + 19.5 = 40$

Anywhere > 40 or -11.5 is an outlier

Here $45 > 40 \rightarrow$ outlier



calculate five number summary and also find outliers using boxplot.

D = [5, 7, 8, 10, 12, 13, 15, 18, 20, 21, 22, 25, 45, 60, 75].

Here! 45, 60 and 75 are suspected outliers.

Step (1) order the data
(already sorted above)

Step (2) Find the median (Q_2)

15 value → median is 8th value.

$$Q_2 = 18$$

Step (3) Find Q_1 (lower quartile)

Lower half = [5, 7, 8, 10, 12, 13, 15].

median of these = 10

$$Q_1 = 10$$

Boxplot descript

Step ④ Find Q3 (upper Quartile)

upper half = [20, 21, 22, 25, 45, 60, 75]

median of this = 25.

$Q_3 = 25.$

Step ⑤ Five-Number Summary

(min, Q1, Q2, Q3, max) = (5, 10, 18, 25, 75)

Step ⑥ IQR and outlier detection

$IQR = Q_3 - Q_1 = 25 - 10 = 15.$

Lower fence $= Q1 - 1.5 * IQR$

$= 10 - 22.5 = -12.5$

Upper fence $= Q_3 + 1.5 * IQR$

$= 25 + 22.5 = 47.5$

Any value >47.5 is an outlier.

outlier = {60, 75}.

45 is not an outlier (since 45<47.5)

Final Boxplot :-

whiskers min = 5, upper whiskers stop at 45 (largest non outlier)

Box: from Q1 = 10 , Q3 = 25.

median line at 18.

outlier: {60, 75}

Example (2)

Data sets (15 values, with negative outl-

[-20, -15, 2, 3, 5, 6, 7, 8, 10, 12, 14, 15, 16, 18, 20]

Here, -20 and -15 look suspicious, let's check formally.

Step (1) Sort (already sorted)

[-20, -15, 2, 3, 5, 6, 7, 8, 10, 12, 14, 15, 16, 18, 20].

Step (2) Find. median (Q_2)

• 15 value $\longrightarrow$ median = 8th value = 8.

$Q_2 = 8$.

Step (3) Find Q1 (lower quartile)

Lower half = [-20, -15, 2, 3, 5, 6, 7]

median = 3

$Q1 = 3$

step(4): Find Q3 (upper Quartile)

upper half = [10, 12, 14, 15, 16, 18, 20]

median = 15

step(5) Five Number Summary:

(min, Q1, Q2, Q3, max) = (-20, 3, 8, 15, 20).

step(6) IQR and outlier detection.

IQR = Q3 - Q1 = 15 - 3 = 12

Lower fence = Q1 - 1.5*IQR = 3 - 18 = -15

Higher fence = Q3 + 1.5*IQR = 15 + 18 = 33

outlier condition: value < -15 or > 33.

So:

outlier = {-20}

-15 is exactly at the lower fence, so it is not the outlier.

Boxplot descriptions

Q1 = 3, Q3 = 15, median = 8,

outlier : -20